

The Pointwise Determinant Weight Does Not Give Exact Nonlinear Loomis–Whitney Constant One

FA.BANACH.001, agent lane 07

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Status: candidate counterexample, likely valid, pending human review.

1 Source question

Carbery, Hänninen, and Valdimarsson prove a sharp weighted nonlinear Loomis–Whitney inequality using a weight $W(x)$ built from ordered fibre flows [1, Proposition 9.2]. Since $W(x) \leq |\omega_1(x) \wedge \cdots \wedge \omega_n(x)|$, they ask on PDF page 48 whether, for sufficiently small V ,

$$\int_V \prod_{j=1}^n f_j(\pi_j x)^{1/(n-1)} |\omega_1(x) \wedge \cdots \wedge \omega_n(x)|^{1/(n-1)} dx \leq \prod_{j=1}^n \left(\int_{U_j} f_j \right)^{1/(n-1)}. \quad (\text{Q})$$

Here $\pi_j : V \rightarrow U_j \subset \mathbb{R}^{n-1}$ are C^1 submersions and ω_j is the wedge product of the rows of $d\pi_j$, identified with a fibre-tangent vector.

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ANTHONY CARBERY, TIMO S. HÄNNINEN AND STEFÁN INGI VALDIMARSSON

Noting that $W(x) \leq \omega_1(x) \wedge \cdots \wedge \omega_n(x)$, one might ask whether

$$\int_V \prod_{j=1}^n f_j(\pi_j x)^{1/(n-1)} \omega_1(x) \wedge \cdots \wedge \omega_n(x)^{1/(n-1)} dx \leq \prod_{j=1}^n \left(\int_{U_j} f_j \right)^{1/(n-1)}$$

holds for sufficiently small V .

We answer (Q) negatively for $n = 3$.

2 Proof intuition

At the origin the proposed weight freezes to the sharp linear Loomis–Whitney normalization, so centred Gaussians are extremal to zeroth order. We perturb all three coordinate projections by a cyclic quadratic “twist.” The first variation cancels by symmetry. At second order the determinant weight gains the Gaussian mean of $|X|^2$, while the nonlinear arguments lose the mean of $X^2 Y^2 + X^2 Z^2 + Y^2 Z^2$. The difference is $3/4 > 0$.

The source asks whether shrinking V can rescue constant one. It cannot: put the Gaussian at scale σ inside any prescribed neighbourhood. The positive excess is of order σ^2 relative to the main integral, whereas the part cut off outside the neighbourhood is exponentially small in $1/\sigma^2$.

3 Counterexample

For $(x, y, z) \in \mathbb{R}^3$, define

$$\begin{aligned}\pi_1(x, y, z) &= (y + xz, z - xy), \\ \pi_2(x, y, z) &= (z + xy, x - yz), \\ \pi_3(x, y, z) &= (x + yz, y - xz).\end{aligned}\tag{1}$$

Let ω_j be the cross product of the two rows of $d\pi_j$, with the displayed coordinate order. Direct differentiation gives

$$\begin{aligned}\omega_1 &= (1 + x^2, -xy - z, y - xz), \\ \omega_2 &= (z - xy, 1 + y^2, -yz - x), \\ \omega_3 &= (-xz - y, x - yz, 1 + z^2).\end{aligned}\tag{2}$$

In particular every π_j is a submersion. Put $J = \det(\omega_1, \omega_2, \omega_3)$. Then

$$\begin{aligned}J(x, y, z) &= 1 + 2(x^2 + y^2 + z^2) \\ &\quad + x^4 + y^4 + z^4 - 2x^2y^2 - 2x^2z^2 - 2y^2z^2 - 4x^2y^2z^2.\end{aligned}\tag{3}$$

Thus $J(0) = 1$, so $J > 0$ on some neighbourhood V_0 of the origin.

Theorem 1. *For the submersions in (1), inequality (Q) fails on every open neighbourhood V satisfying $0 \in V \subset V_0$. Consequently there is no “sufficiently small” neighbourhood on which the proposed exact determinant-weight inequality holds for all nonnegative inputs.*

Proof. Fix such a V and choose $r > 0$ with $B(0, r) \subset V$. Set $U_j = \pi_j(V)$ and, for $\sigma > 0$, let

$$f_{j,\sigma}(u) = e^{-|u|^2/\sigma^2} \quad (u \in U_j).\tag{4}$$

The right side of (Q) is at most

$$\prod_{j=1}^3 \left(\int_{\mathbb{R}^2} e^{-|u|^2/\sigma^2} du \right)^{1/2} = \pi^{3/2} \sigma^3.\tag{5}$$

We first evaluate the corresponding full-space integral with $|J|^{1/2}$. Write $x = \sigma X$, $y = \sigma Y$, $z = \sigma Z$, and put

$$R^2 = X^2 + Y^2 + Z^2, \quad S_4 = X^2Y^2 + X^2Z^2 + Y^2Z^2.$$

The cyclic signs in (1) make all linear cross terms cancel, giving the exact identity

$$\sum_{j=1}^3 |\pi_j(x, y, z)|^2 = 2(x^2 + y^2 + z^2) + 2(x^2y^2 + x^2z^2 + y^2z^2).\tag{6}$$

Moreover, (3) gives

$$\begin{aligned}J(\sigma X, \sigma Y, \sigma Z) &= 1 + 2\sigma^2 R^2 \\ &\quad + \sigma^4 (X^4 + Y^4 + Z^4 - 2X^2Y^2 - 2X^2Z^2 - 2Y^2Z^2) - 4\sigma^6 X^2Y^2Z^2.\end{aligned}\tag{7}$$

Hence, pointwise on bounded sets,

$$e^{-\sigma^2 S_4} |J(\sigma X, \sigma Y, \sigma Z)|^{1/2} = 1 + \sigma^2 (R^2 - S_4) + O(\sigma^4 P(X, Y, Z)),\tag{8}$$

where P is a fixed polynomial.

For completeness, (8) may be integrated term by term. Split at $R = \sigma^{-1/4}$. On the inner region $J > 0$ and Taylor's formula gives an integrable polynomial remainder after multiplication by e^{-R^2} . On the outer region the Gaussian tail is $O(e^{-\sigma^{-1/2}/2})$, even after the polynomial determinant factor. Therefore

$$\begin{aligned}\mathcal{I}_\sigma &:= \int_{\mathbb{R}^3} \prod_{j=1}^3 e^{-|\pi_j(x)|^2/(2\sigma^2)} |J(x)|^{1/2} dx \\ &= \sigma^3 \int_{\mathbb{R}^3} e^{-R^2} [1 + \sigma^2(R^2 - S_4) + O(\sigma^4 P)] dX dY dZ.\end{aligned}\tag{9}$$

Under the probability density $\pi^{-3/2}e^{-R^2}$, the coordinates are independent, have second moment $1/2$, and satisfy

$$\mathbb{E}R^2 = \frac{3}{2}, \quad \mathbb{E}S_4 = 3 \left(\frac{1}{2}\right)^2 = \frac{3}{4}.\tag{10}$$

It follows that

$$\mathcal{I}_\sigma = \pi^{3/2} \sigma^3 \left(1 + \frac{3}{4}\sigma^2 + O(\sigma^4)\right).\tag{11}$$

It remains only to localize. Identity (6) implies that the exponential in $\mathcal{I}_\sigma$ is bounded above by $e^{-(x^2+y^2+z^2)/\sigma^2}$, while (3) gives $|J(x)|^{1/2} \leq C(1 + |x|^3)$. Consequently

$$\int_{\mathbb{R}^3 \setminus B(0,r)} \prod_{j=1}^3 e^{-|\pi_j(x)|^2/(2\sigma^2)} |J(x)|^{1/2} dx = O(e^{-r^2/(2\sigma^2)}) = o(\sigma^5).\tag{12}$$

Because $J > 0$ on V , equations (11)–(12) show that the left side of (Q), with the functions (4), is strictly larger than $\pi^{3/2}\sigma^3$ for all sufficiently small σ . This contradicts the upper bound (5) and proves the theorem. $\square$

4 Compatibility with the source setup

The vectors in (2) never vanish, so each one generates the local C^1 fibre flow required in the source. At the origin they are the three coordinate vectors. The ordered flow-composition map therefore has identity derivative in the flow parameters at the origin. After shrinking V_0 , its derivative is uniformly within (say) $1/2$ of that identity on a convex parameter box; the mean-value estimate then makes the composition injective. Thus the example fits the local hypotheses preceding the source question, not merely the displayed algebraic inequality.

5 Verifier report

The proof was checked in the following ways.

1. Direct differentiation verifies the three fibre vectors (2) and shows the submersion condition globally.
2. Expanding their determinant gives (3), with $J(0) = 1$.
3. Expanding the squared maps gives the positive identity (6); this also supplies the Gaussian tail needed for localization.
4. The exact symbolic checker in `code/check_symbolic_expansion.py` independently verifies (2), (3), (6), the second-order polynomial in (8), and its Gaussian mean $3/4$.

5. The computational second-variation search used to discover the cyclic twist is not used as proof. The formal argument rests only on the displayed polynomial identities, Taylor expansion, Gaussian moments, and tail bound.

Recommended human-review focus: check that the source intends the pointwise wedge magnitude in (Q), and audit the uniform Taylor/tail split between (8) and (12).

6 Limitations and novelty status

The counterexample concerns the proposed replacement of $W(x)$ by the larger pointwise determinant. It does not contradict Proposition 9.2, Corollary 9.3, or the standard $(1 + \varepsilon)$ nonlinear Brascamp–Lieb theorem. It also does not supply the unrelated closed-form sharp factorisations discussed later in the source.

A bounded search on 19 July 2026 used arXiv:1809.02449, the exact title and authors, and the phrases “nonlinear Loomis–Whitney,” “determinant weight,” “sharp constant,” and “counterexample.” It also checked the announced sequel arXiv:2003.03326 and later nonlinear Brascamp–Lieb records. No later resolution or matching counterexample to the exact page-48 question was found. This is a bounded novelty check, not an exhaustive priority claim.

References

- [1] A. Carbery, T. S. Hänninen, and S. I. Valdimarsson, *Multilinear duality and factorisation for Brascamp–Lieb-type inequalities with applications*, J. Eur. Math. Soc. 25 (2023), 2057–2125; arXiv:1809.02449.